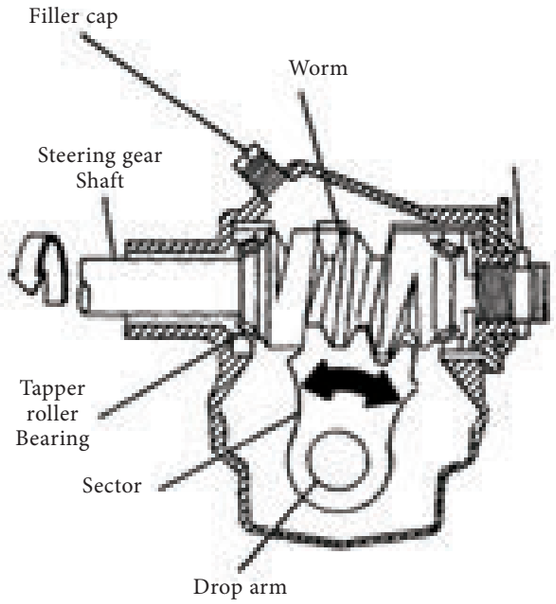


Figure 8.3.1.3 Worm and sector steering box

the steering wheel is converted into angular motion of the sector gear.

- The drop arm attached to the sector is also oscillated and through the steering linkages, the wheel is turned to the required degrees in the required direction.

8.3.1.4 Recirculating ball type steering gear box

It consists of a worm at the end of a steering rod. A nut is mounted on the worm with two sets of balls in the grooves of the worm, in between the nut and the worm. The balls reduce the friction during the movement of the nut on the worm. The nut has a number of teeth on the outside, which mesh with the teeth on a worm wheel sector, on which is further mounted the drop arm. When the steering wheel is turned, the balls in the worm roll in the grooves and cause the nut to travel along the length of the worm. The balls, which are in 2 sets, are recirculated through the guides. The movement of the nut causes the wheel sector to turn at an angle and actuate the link rod through the drop arm, resulting in the desired steering of the wheels.

8.4 Power steering

- In medium and heavy commercial vehicles, the driver has to put lot of physical effort to turn the vehicle. If the driver continuously drives the vehicle for longer duration, then the driver will be very tired.
- In order to reduce the steering effort by the driver for easy turning, an energy source that aids the driver in turning the wheels is called as power

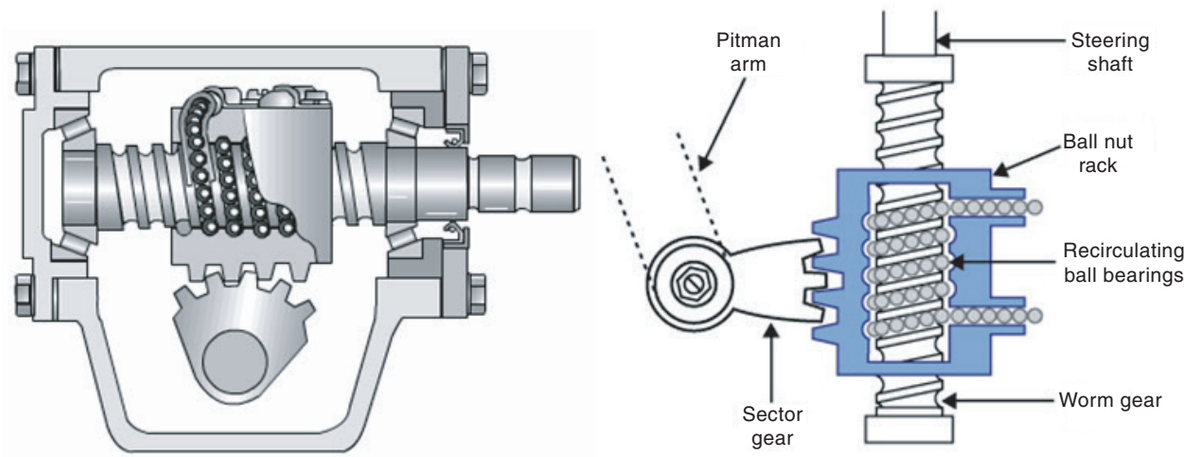


Figure 8.3.1.4 Recirculating ball type steering gear box

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